Worksheet 9: probabilistic forecasting

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Brier score $= \frac{1}{N} \sum_i (p_i - y_i)^2$

$$y_i = \begin{cases} 0 & \text{no rain} \\ 1 & \text{rain} \end{cases}$$

Days	Forecast	Obs. y_i	$(p_i - y_i)^2$
1	0.2	0	0.04
2	0.7	1	0.09
3	0.1	0	0.01
4	0.9	1	0.01
5	0.4	0	0.16

$$BS = \frac{0.31}{5} = 0.06$$

You could compare this with another forecast model to see which is better.

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$$
\begin{aligned}
LS(\mu, \sigma) &= -\log f(y) \\
&= +\frac{1}{2}\log(2\pi\sigma^2) + \frac{(y-\mu)^2}{2\sigma^2}
\end{aligned}
$$

Suppose observation $y = 12$ and forecast $\mu = 10, \sigma = 3$

$$
\begin{aligned}
LS &= \frac{1}{2}\log(2\pi 3^2) + \frac{(12-10)^2}{2 \times 3^2} \\
&= 2.24
\end{aligned}
$$

Again $\rightarrow$ compare with another forecast.
We would average these over multiple forecasts.

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$CRPS(\mu, \sigma, y) = \sigma\Big[z(2\Phi(z) - 1) + 2\phi(z) - \frac{1}{\sqrt{\pi}}\Big]$

where Φ is the normal cdf and ϕ is the pdf.

$z = \frac{x - \mu}{\sigma}$

Observation $y = 12, \mu = 10, \sigma = 3$
$z = \frac{12 - 10}{3} = \frac{2}{3}$

$\Phi(z) = 0.7475, \quad \phi(z) = 0.3194$

$CRPS = 3\Big[\frac{2}{3}(2 \times 0.7475 - 1) + 2 \times 0.3194 - \frac{1}{\sqrt{\pi}}\Big]$
$= 1.214$

from scipy.stats import norm
norm.cdf
norm.pdf

1 The General Energy Form

The CRPS is often written as:

$$CRPS(F, y) = \mathbb{E}[|X - y|] - \frac{1}{2}\mathbb{E}[|X - X'|]$$

But this is a bit abstract. How do we actually calculate it?

For a finite ensemble of M members denoted by $\{x_1, x_2, \ldots, x_M\}$ and a single scalar observation y, the CRPS is calculated as follows:

$$CRPS(y, \{x_1, \ldots, x_M\}) = \frac{1}{M}\sum_{i=1}^{M} |x_i - y| - \frac{1}{2M^2}\sum_{i=1}^{M}\sum_{j=1}^{M} |x_i - x_j|$$

- Reliability term: the first part of the equation measures the mean absolute error of the ensemble members relative to the truth
- Uncertainty term: the second part measures the internal spread or diversity of the forecast members
- Interpretation: a lower score indicates a more accurate and appropriately confident forecast

2 Worked Numerical Example

Consider a toy data set for a 2-meter temperature forecast at a specific grid point.

- Observation (y): 10
- Ensemble members ($M = 3$): $\{8, 11, 14\}$

“

2.1 Step 1: The reliability term

Calculate the absolute difference between each member and the observation:

- $|8 - 10| = 2$
- $|11 - 10| = 1$
- $|14 - 10| = 4$

The mean of these differences is:

$$\text{Reliability} = \frac{2 + 1 + 4}{3} = \frac{7}{3} \approx 2.333$$

	8	**11**	**14**
8	0	3	6
11	3	0	3
14	6	3	0

2.2 Step 2: The uncertainty term

Calculate the absolute differences between all pairs of members $|x_i - x_j|$. This can be visualized as a matrix:

The sum of all elements in this matrix is $3 + 6 + 3 + 3 + 6 + 3 = 24$. We divide this by $2M^2$:

$$\text{Uncertainty} = \frac{24}{2 \times 3^2} = \frac{24}{18} \approx 1.333$$

2.3 Step 3: Final result

$$CRPS = 2.333 - 1.333 = 1.000$$

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$$\begin{bmatrix} \mu \\ \sigma \end{bmatrix} = \begin{bmatrix} 0.32 \\ -0.242 \end{bmatrix}$$

so $\mu = 0.32$

$\sigma = \text{softplus}(-0.242)$
$= \log(1 + e^{-0.242})$
$= 0.251$

Note $\sigma = \log(1 + e^{s})$
$e^{\sigma} = e^{s} - 1$
$s = \log(e^{\sigma} - 1)$

Suppose $x = \begin{bmatrix} 0.5 & -0.2 \end{bmatrix}$

$$W = \begin{bmatrix} 0.8 & -0.4 \\ 0.3 & 0.6 \end{bmatrix}, \quad b = \begin{bmatrix} 0.1 \\ -0.05 \end{bmatrix}$$

$$z = Wx + b = \begin{bmatrix} 0.8 & -0.4 \\ 0.3 & 0.6 \end{bmatrix} \begin{bmatrix} 0.5 \\ -0.2 \end{bmatrix} + \begin{bmatrix} 0.1 \\ -0.05 \end{bmatrix} = \begin{bmatrix} 0.58 \\ -0.02 \end{bmatrix}$$

$$h = \text{ReLU} \begin{bmatrix} 0.58 \\ -0.02 \end{bmatrix} = \begin{bmatrix} 0.58 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} \mu \\ \sigma \end{bmatrix} = W_y h + b = \begin{bmatrix} 0.5 & -0.2 \\ 0.1 & 0.3 \end{bmatrix} \begin{bmatrix} 0.58 \\ 0 \end{bmatrix} + \begin{bmatrix} 0.03 \\ -0.3 \end{bmatrix} = \begin{bmatrix} 0.32 \\ -0.242 \end{bmatrix}$$

The PIT histogram

The Probability Integral Transform (PIT) is used to evaluate the calibration of a probabilistic forecast. For a perfectly calibrated ensemble, the observation y should be indistinguishable from the ensemble members. This results in a uniform distribution of the observation's rank within the sorted ensemble.

3 Numerical Example

Let us assume we have an ensemble of size $M = 3$. We will evaluate three different days to see where the observations fall.

3.1 Case 1: Day One

- Sorted ensemble: $\{10, 12, 15\}$
- Observation y: 11
- Since $10 \leq 11 < 12$, the Rank $R = 2$

3.2 Case 2: Day Two

- Sorted ensemble: $\{9, 11, 14\}$
- Observation y: 16
- Since $16 > 14$, the Rank $R = 4$

3.3 Case 3: Day Three

- Sorted ensemble: $\{10, 13, 16\}$
- Observation y: 8
- Since $8 < 10$, the Rank $R = 1$

4 Constructing the Histogram

After these three days, our set of ranks is $\{2, 4, 1\}$.

- We create a histogram with $M + 1$ bins (in this case, 4 bins). We often standardise with $\tilde{R} = R/(M + 1)$
- We count how many times each rank occurs
- A perfectly calibrated model would result in each bin having approximately the same number of counts after many trials